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On the nature of T($Al_2Mg_3Zn_3$) and S(Al_2CuMg) phases present in as-cast and annealed 7055 aluminum alloy

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Abstract

Aluminum alloys, encompassed by AA 7055 alloy composition, having the nominal zinc content (i.e. 8 wt.%) but varying copper and magnesium contents across the alloy composition range were examined in the as-cast form by a combination of light microscopy, scanning electron microscopy (SEM), electron probe micro analysis (EPMA) and X-ray diffraction (XRD). It is observed that for all compositions, the second phases based on η($MgZn_2$), T($Al_2Mg_3Zn_3$) and S(Al_2CuMg) are present. The T phase dissolves copper up to 28 wt.%, whilst the S phase shows metastable solubility of zinc that may range up to 30 wt.%. In alloys with magnesium at the lower limit and the copper contents approaching the upper limit of the alloy composition, the θ phase (Al_2Cu) of the constituent binary Al–Cu system is further observed. The θ phase (Al_2Cu) does not dissolve either zinc or magnesium. Below the nominal composition, the alloys could be homogenized substantially using a commercially viable homogenization treatment leaving small amounts of undissolved S phase that does not contain any zinc.
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Keywords: Aluminum alloy 7055; Ingot casting; Annealing; T($Al_2Mg_3Zn_3$) and S(Al_2CuMg) phases; Electron probe microanalysis; X-ray diffraction

1. Introduction

In the past several decades, considerable commercial interest has been shown in the development of high strength Al–Zn–Mg–Cu base 7XXX series aluminum alloys aiming at a superior combination of high specific strength, fracture toughness and stress corrosion cracking resistance that renders them suitable for use predominantly in aircraft structures [1–5]. This led to the design and development of the Al–Zn–Mg–Cu–Zr base AA 7055 alloy produced by Alcoa in the early 1990's. This high solute alloy evokes the highest strength aluminum alloy produced by ingot metallurgical route and finds application as upper wing structural material in the state-of-the-art Boeing 777 aircraft [6]. It is well known that the superior combination of properties promised by this alloy is largely determined by the stability, morphology as well as the chemistry of the strengthening precipitates, the dispersiods and the various constituent (insoluble) phases present in the alloy. The nature of the secondary alloy phases present in this complex alloy system is not fully understood in that the chemistry of many of the secondary phases in the constituent binary and ternary systems [such as Al–Cu, Al–Cu–Mg, Al–Zn–Mg, etc.] may be modified when such phases form in multi-component alloy systems. Such modifications in the nature of the secondary alloy phases may directly affect both mechanical properties and corrosion behavior of the alloy.

Existing literature suggests that below solidus temperature, four major intermetallic phases, viz. η($MgZn_2$), T($Al_2Mg_3Zn_3$), S(Al_2CuMg) and θ(Al_2Cu) can occur in Al–Zn–Mg–Cu alloys [3,5]. Whilst, η and T phases have extended solubility of copper in them, θ and S are essentially binary and ternary phases in the constituent Al–Cu and Al–Cu–Mg systems, respectively [5]. The Zn:Mg ratio plays a crucial role in controlling the nature of zinc-bearing constituents. Copper behaves synchronously with zinc in this respect, and up to 2.5 wt.% copper additions, most of the copper remains dissolved in the above two intermetallic phases

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i.e. η and T [5]. Above 2.5 wt.%, however, copper tends to form S phase rather than dissolving in the matrix or in η and T phases [5]. Furthermore, eutectic phases involving η dissolve fast during homogenization, whilst, the dissolution rate of T and S phases is rather sluggish [7]. Given the varying nature of the secondary alloy phases (in terms of their compositions and their response to homogenizing annealing process), it is important that the chemistry of the coarse intermetallic phases present in the as-cast form of these alloys be characterized in details.

Studies have further shown that the chemistry of the coarse, equilibrium alloy phases present in the as-cast microstructure is similar to that of the same phases when formed as fine precipitates during aging following solution treatment and quenching [8–10]. Such studies demonstrated that the as-cast alloy may often contain metastable phases, and studies involving such phases may provide a basis for further understanding of the solid-state phase transformations that occur in these alloys [8–10]. Evaluation of the exact chemistry of the second phases, when present as coarse particles in the as-cast and/or annealed microstructure, also becomes much easier using conventional micro-analytical techniques [8–10].

The objective of the present study is to provide an understanding of the nature of secondary alloy phases present in as-cast and subsequently annealed microstructure of 7055 aluminum alloy with a particular attention to T and S phases.

2. Experimental procedure

7055 aluminum alloys having the average zinc content (i.e. 8 wt.%), zirconium content of 0.16 wt.% and varying magnesium and copper contents across the composition range (1.8–2.3 wt.% Mg and 2.0–2.6 wt.% Cu) were prepared via ingot metallurgical route. The alloy samples were taken from the equivalent position of the moulds in order to maintain uniformity of cooling rate. For each composition, 15 kg of the alloy was cast in a bottom pouring, tapered mild steel mould of dimension 310 mm × 310 mm × (50–40) mm. After casting, a slice of as-cast alloy of dimension 310 mm × 10 mm × (50–40) mm was machined from the surface of each ingot. All the specimens were collected from the middle of such slices, where the average local cooling rate [in the temperature interval of 710 °C i.e. the pouring temperature of the alloy to 200 °C] was measured to be ∼15 °C/min. The cooling rate was calculated from an experimental temperature-time plot obtained by placing a suitable thermocouple close to the said region.

The mass compositions of the alloys examined in the present investigation (designated A through E) are given in Table 1. The alloys were given a commercially viable homogenizing annealing treatment at 450 °C for 35 h. The homogenization temperature was selected on the basis of results of thermal analysis (by differential scanning calorimetry) of as-cast, high zinc containing (>8 wt.%) Al–Zn–Mg–Cu–Zr alloy showing the presence of non-equilibrium solidus at 460 °C [11].

Table 1
Compositions (wt.%) of the 7055 Aluminum Alloys used in the Investigation

Alloy	Zn	Mg	Cu	Zr	Fe	Si	Al
A	8.0	2.3	2.6	0.16	0.06	0.05	Balance
B	8.0	2.0	2.3	0.16	0.04	0.05	Balance
C	8.0	1.8	2.6	0.16	0.06	0.06	Balance
D	8.0	1.8	2.0	0.16	0.06	0.05	Balance
E	8.0	2.3	2.0	0.16	0.05	0.06	Balance

A combination of optical microscopy, scanning electron microscopy (SEM) and electron probe micro-analysis (EPMA) was carried out to characterize as-cast and annealed microstructures of the alloys. SEM, in the back scattered electron imaging mode, was carried out on the SEM attached to the EPMA instrument, operating at 20 kV. EPMA was performed on a CAMECA CAMEBAX micro-analyzer operating at 20 kV. A quantitative X-ray wavelength dispersive spectroscopy (WDS) system attached to the EPMA instrument was used to analyze the chemistry of the phase particles present in as-cast and annealed microstructures. The instrument was equipped with the MAGIC-IV analytical program for ZAF corrections [12]. For each type of phase, at least five particles from different locations were analyzed and the average values of such analyses are reported here. The analyses were carried out on polished but unetched samples. X-ray diffraction studies (XRD) were undertaken in order to identify the alloy phases present in as-cast and annealed materials. XRD was further used to measure the lattice parameter of certain alloy phases; alloy E (see Table 1) was utilized only for this purpose. X-ray diffraction studies were performed on a PHILIPS PW 1710 automatic diffractometer.

3. Results and discussion

3.1. General description of microstructure

For all the alloys, optical microscopy and SEM revealed a typical cored dendritic microstructure of primary α(Al)-solid solution surrounded by inter-dendritic secondary phases. The predominant eutectic structure of the alloys is understood to be consisting of a 'quasi-binary reaction' products, evolving from parallel solidification of three quasi-binary eutectic reactions viz. α-Al/η[$MgZn_2$], α-Al/T[$Al_2Mg_3Zn_3$] and α-Al/S[Al_2CuMg] [7]. Apart from being a part of eutectic structure, T-base and S-base phases are also present as '*divorced*' phases. Compared to η- and T-base phases, the S-base phase is present in relatively smaller volumes, although the volume fraction of S phase is found to increase with increasing copper and magnesium contents of the alloys. The θ(Al_2Cu) phase of the constituent Al–Cu system is also observed in alloys with magnesium toward the lower limit and copper contents approaching the upper limit of the composition range.

Fig. 1(a) through (d) represents optical micrographs showing as-cast microstructures of 7055 aluminum alloys having

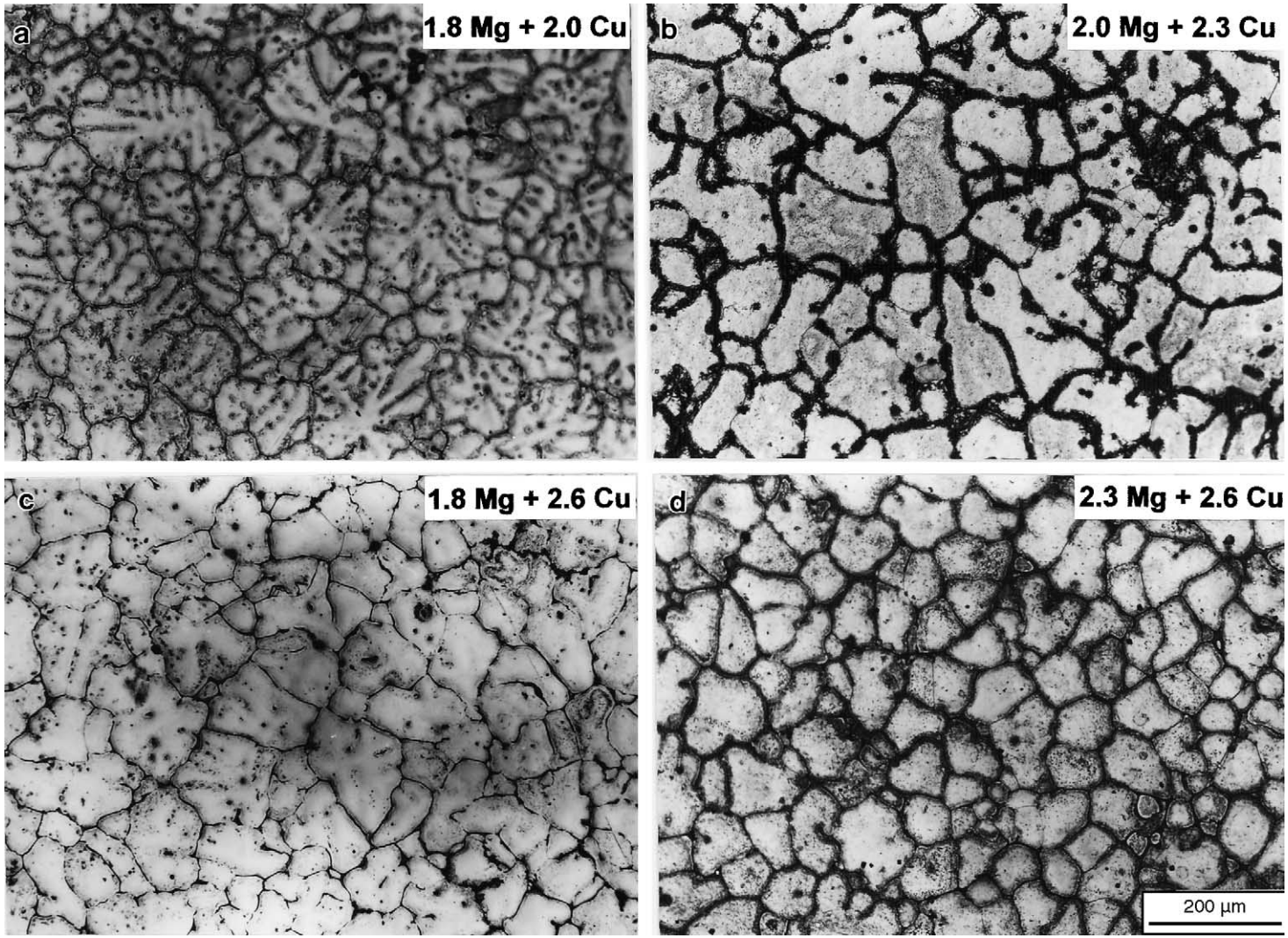

Fig. 1. Optical micrographs of as-cast 7055 Al alloys showing cored dendritic microstructures. The Mg and Cu contents of the alloys are labeled in the micrographs.

varying magnesium and copper contents. Gradual refinement of as-cast grain size with increasing amount of solutes is evident in Fig. 1. This parallels a recent study where magnesium additions were found to exert a similar effect on as-cast grain morphology of an Al–Zn–Cu–Zr alloy [13].

3.2. Secondary alloy phases present in as-cast materials

In alloys A, B and C, the second phases identified are based on $\eta(MgZn_2)$, $T(Al_2Mg_3Zn_3)$, $\theta(Al_2Cu)$ and $S(Al_2CuMg)$ phases. Except the θ phase, other phases could be detected conclusively in alloy D. The XRD traces of as-cast alloys are shown in Fig. 2. The XRD peaks could be identified due to α(Al) solid solution (labeled Al), $MgZn_2$ (labeled η), $Al_2Mg_3Zn_3$ (labeled T), Al_2CuMg (labeled S), and Al_2Cu (labeled θ); the corresponding crystallographic planes are also marked in parentheses. Fig. 2 shows that the presence of $\theta(Al_2Cu)$ phase was detected by XRD only in alloy C having the highest Cu:Mg ratio, whilst in other two alloys (i.e. A and B) the θ phase could be identified only by EPMA. This may

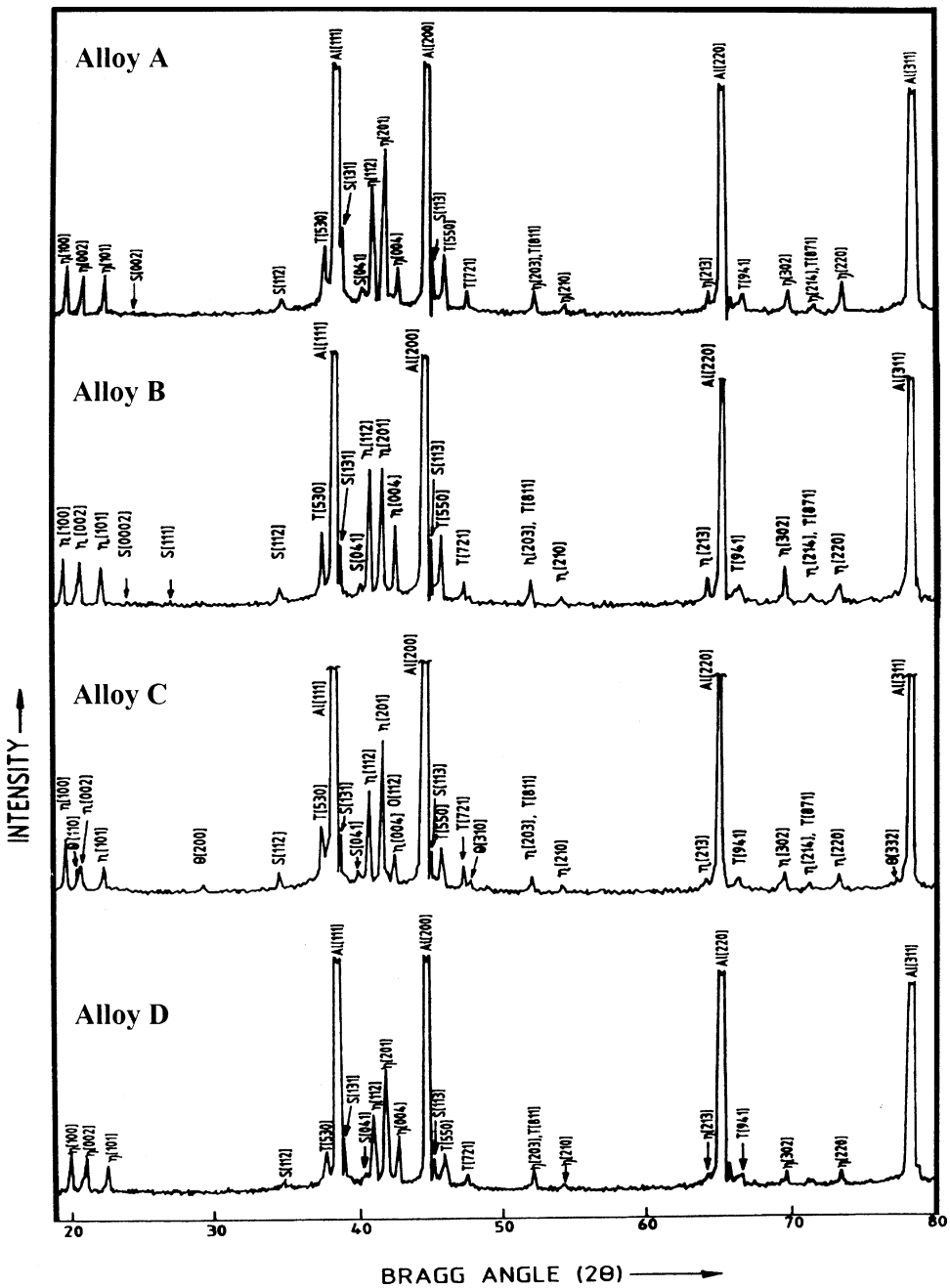

Fig. 2. XRD traces of as-cast 7055 aluminum alloys having 8 wt.% zinc, 0.16 wt.% zirconium and varying copper and magnesium contents (wt.%) i.e. alloy A containing 2.3 wt.% Mg and 2.6 wt.% Cu, alloy B containing 2.0 wt.% Mg and 2.3 wt.% Cu, alloy C containing 1.8 wt.% Mg and 2.6 wt.% Cu, and alloy D containing 1.8 wt.% Mg and 2.0 wt.% Cu.

be due to very low volumes of θ phase present in alloys A and B.

In any alloy, the dominant second phase is based on η($MgZn_2$), present in the form of a eutectic structure. The next dominant phase is based on T($Al_2Mg_3Zn_3$), which is present mostly as divorced phase (also as a part of the eutectic). S(Al_2CuMg) phase is found to be present in rather small volumes, mostly as part of the eutectic.

Fig. 3(a) represents a scanning electron micrograph (BSE image) showing typical as-cast microstructure of alloy B. The presence of coarse secondary phases at the inter-dendritic regions is evident in the micrograph. Quantitative WDS analyses of the phase particles indicated the presence of four different types of particles namely, η, T, S and θ phases. The chemical composition of the η phase could not be determined accurately as the η phase is almost always present as a part of the eutectic structure. However, a defocused beam WDS analysis [that gives an average composition of α-Al + η eutectic] shows that the eutectic phase mixture contains 16–18 wt.% copper, whereas, copper content of the matrix (α-Al) is only about 1–1.3 wt.%. This definitely indicates that η phase dissolves considerable amounts of copper in it.

Contrary to this, the presence of coarse '*divorced*' phases enabled chemical composition to be determined accurately for the other secondary alloy phases. Fig. 3(b) represents a magnified image of '*divorced*' T phase marked by an arrow in Fig. 3(a). Quantitative WDS analysis shows that the chemical composition of T phase follows the range: 35–38 wt.% zinc, 20–23 wt.% magnesium, 27–28 wt.% copper and the balance being aluminum. Significant amount of copper content (27–28 wt.%) is a noticeable feature of the T phase. A copper line-scan trace of a T phase particle [shown in Fig. 3(b)] reconfirms the presence of copper within the T phase [see Fig. 3(c)].

Table 2 shows the chemical compositions of the secondary alloy phases present in the as-cast alloys. The data show that the copper dissolved in the T phase substitutes for zinc atoms in the phase structure [i.e. (Cu + Zn) wt.% accounts for total

Table 2
Chemical compositions (wt.%) of alloy phases present in as-cast alloys

Alloy	Phases present	Zn	Mg	Cu	Al
A	T-base	38.08	23.1	27.2	11.62
	S-base	24.6	14.2	23.1	38.1
	θ	2.30	0.53	47.75	49.42
B	T-base	35.8	20.5	28.1	15.6
	S-base	29.1	15.2	18.7	37.0
	θ	3.9	0.8	45.7	49.6
C	T-base	35.3	20.1	27.2	17.4
	S-base	25.3	12.9	17.2	44.6
	θ	2.1	0.40	48.32	49.18
D	T-base	36.6	21.5	28.0	13.9
	S-base	30.2	13.6	18.3	37.9
E	T-base	41.4	22.1	23.2	13.3
	S-base	27.9	15.2	18.4	38.5

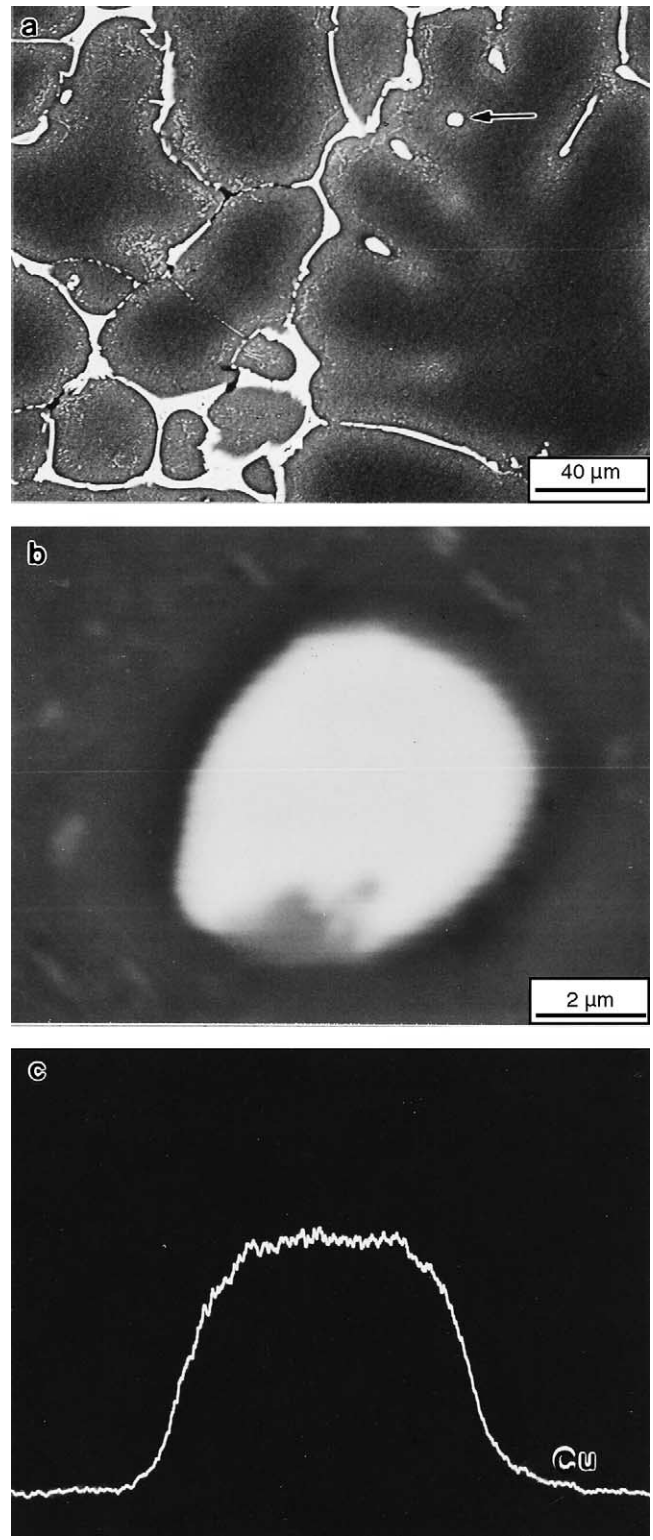

Fig. 3. (a) BSE image of as-cast alloy B, (b) a magnified image of a T phase marked by an arrow in (a). (c) Cu line-scan trace of the T phase [shown in (b)] revealing its association with Cu.

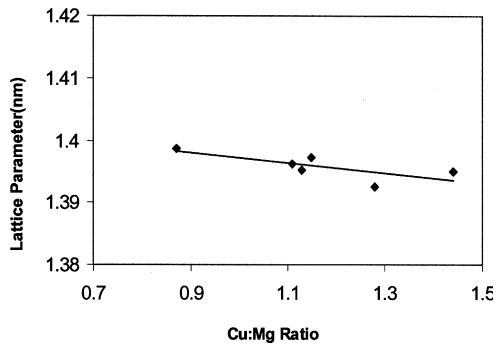

Fig. 4. A plot of lattice parameter (nm) of T phase present in the as-cast alloys vs. Cu:Mg ratio of the alloys. The plot shows decreasing lattice parameter with increasing Cu:Mg ratio.

zinc wt.% in the ternary T phase in the constituent Al–Zn–Mg system]. Hence, it is quite appropriate to designate the T phase as $Al_2Mg_3Zn_3Cu_{3-x}$. Table 2 further shows that increasing Cu:Mg ratio in the alloy composition has the effect of increasing the copper content and reducing the magnesium content in the T phase. Our XRD studies showed that increasing Cu:Mg ratios decrease the lattice parameter of the T phase. Fig. 4, representing a graph of lattice parameter of T phase versus Cu:Mg ratios of 7055 aluminum alloys, depicts such an effect. This is consistent with copper having smaller and magnesium having larger atomic radii than those of zinc and aluminum. Fig. 4 shows that the lattice parameter of cubic T phase decreases from $a = 1.3987$ nm in alloy D to $a = 1.395$ nm in alloy A. This data may be compared with the reported lattice parameter of the cubic T phase, in the ternary Al–Zn–Mg system, ranging from $a = 1.415$ to 1.47 nm [5].

The occurrence of '*divorced*' T phase was often observed to be associated with θ phase. Typically T phase is found at the periphery of θ phase in the alloys with magnesium at the lower limit and copper contents approaching the upper limit of the alloy composition (e.g. alloy C). Fig. 5(a) and (b) represent back-scattered electron images obtained from alloy C showing typical occurrence of T phase around the θ phase particles. Elemental line scan traces of magnesium, copper and zinc shown in Fig. 5(c) and (d) provide the basis of such interpretations. It is suggested that magnesium and zinc are initially associated with θ phase implying metastable solubility of these elements in the θ phase. Rejection of solutes from the θ phase (thus implying insolubility of both magnesium and zinc in θ phase) may cause the formation of T phase at the periphery of the θ phase. This derives support from the compositional analysis that the θ phase dissolves neither zinc nor magnesium [see Table 2].

Fig. 6(a) represents scanning electron micrographs of as-cast alloy B showing the presence of S phase. One such S particle, having typically a circular morphology, is marked by an arrow in Fig. 6(a). Fig. 6(b) shows the magnified image of the S particle. Apart from being part of eutectic, the S phase is also found to be present as a 'divorced' phase having circular as well as irregular shapes. Similar to η and T phases, $S(Al_2CuMg)$ phase is also present in all the as-cast

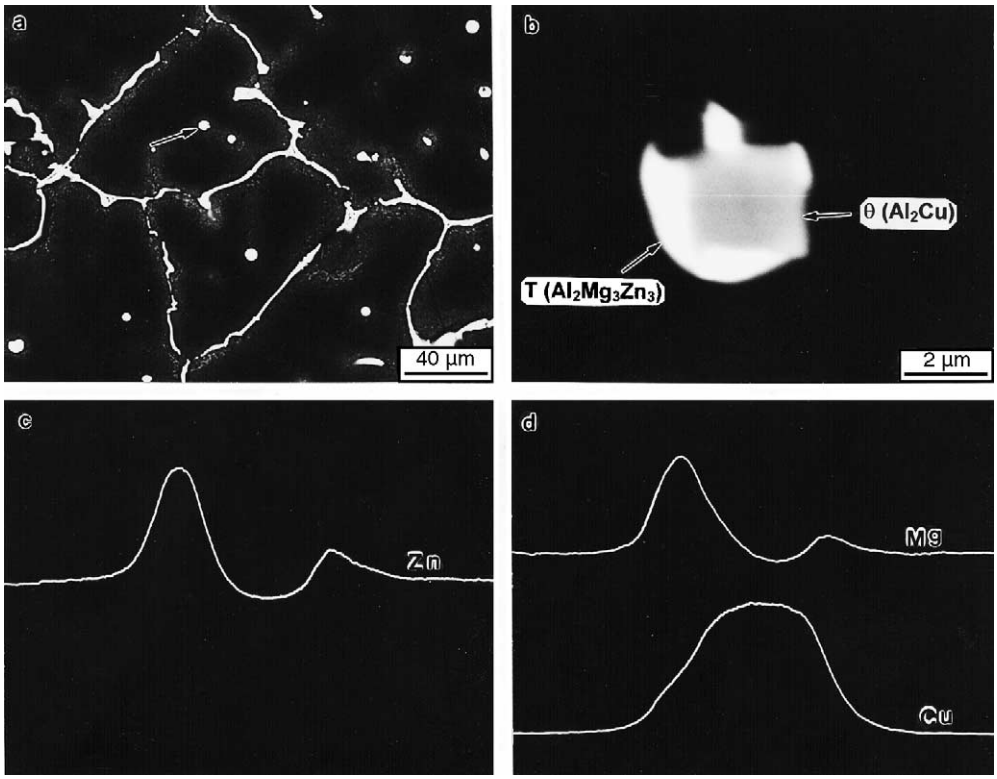

Fig. 5. (a) BSE image of as-cast alloy C, (b) BSE image showing "association of T phase with the θ phase" marked by an arrow in (a). (c) Zn and (d) Mg and Cu line-scan traces of the T–θ phase association shown in (b).

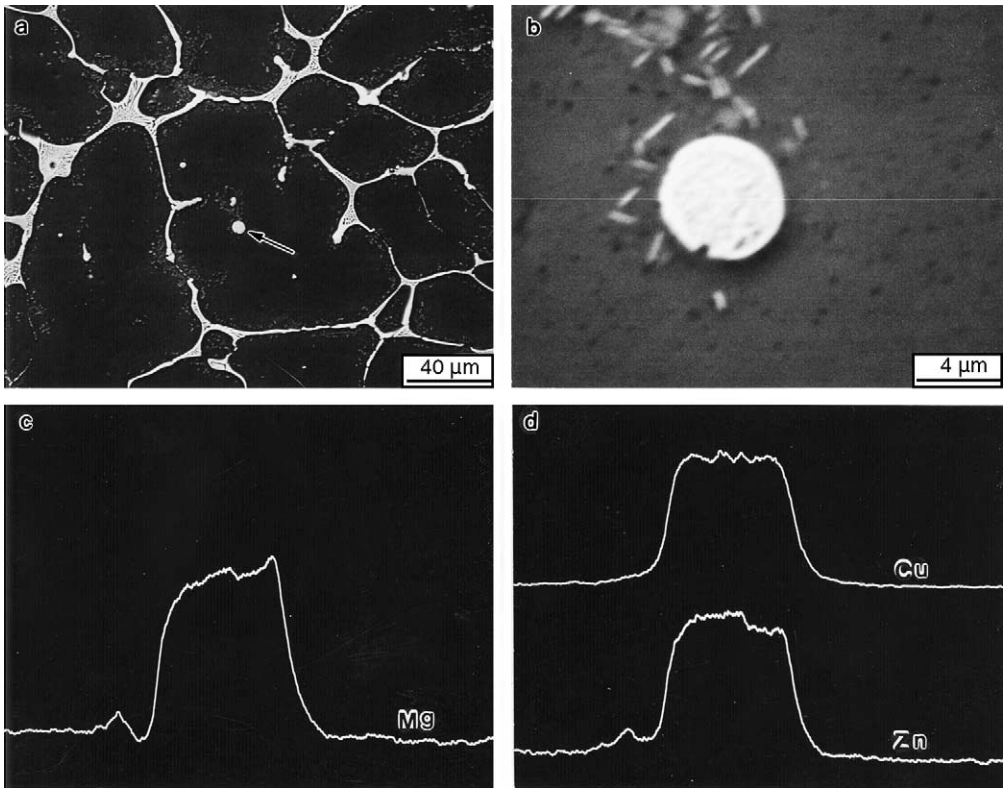

Fig. 6. (a) BSE image of as-cast alloy B, (b) BSE image showing a magnified image of a S phase particle shown in (a). (c) Mg and (d) Cu and Zn line-scan traces of the S phase revealing the association of Zn with the S phase.

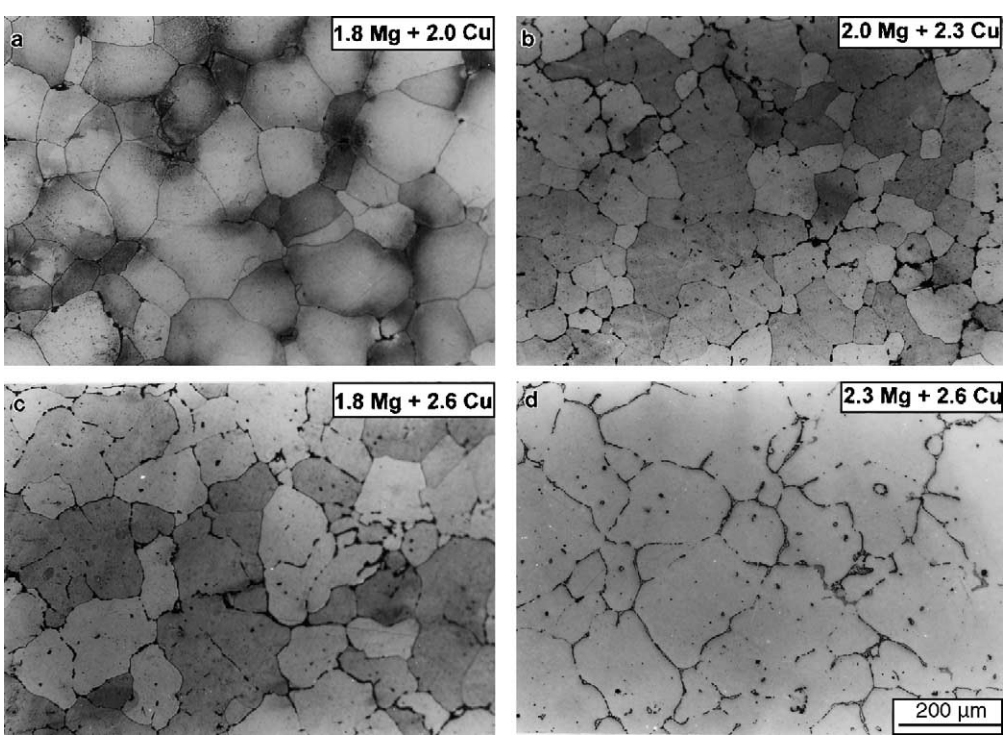

Fig. 7. (a)–(d) Optical micrographs of 7055 aluminum alloys [Cu and Mg contents (wt.%) of each alloy are labeled] showing extent of homogenization after annealing at 450 °C for 35 h.

alloys regardless of composition. As the (Cu + Mg) content of the alloys increases, an increased amount of S phase is observed in the microstructure. Alloy A having the highest (Cu + Mg) content (i.e. 4.9 wt.%) shows the maximum volume of S phase present in the as-cast microstructure. Alloys B and C having similar copper and magnesium contents (i.e. 4.3 and 4.4 wt.%, respectively) show more or less equal S phase contents in the microstructure. Typical elemental line-scan traces from one S phase particle present in alloy B show that the S phase is associated with zinc [Fig. 6(c)]. Table 2 shows that the zinc content in the S phase may vary from 24 to 30 wt.%, and that zinc substitutes for copper in the S phase structure.

3.3. Secondary alloy phases present in annealed materials

Fig. 7(a) through (d) represents light micrographs showing annealed microstructure (35 h at 450 °C) of alloys A through D, respectively. Fig. 7(a) through (d) show that 7055 aluminum alloys having less than the average copper and magnesium contents in the alloy composition could be homogenized '*substantially*' using the commercially viable homogenizing annealing treatment. On the other hand, alloys having higher than and/or equal to the nominal composition of copper and magnesium could not be homogenized to that extent.

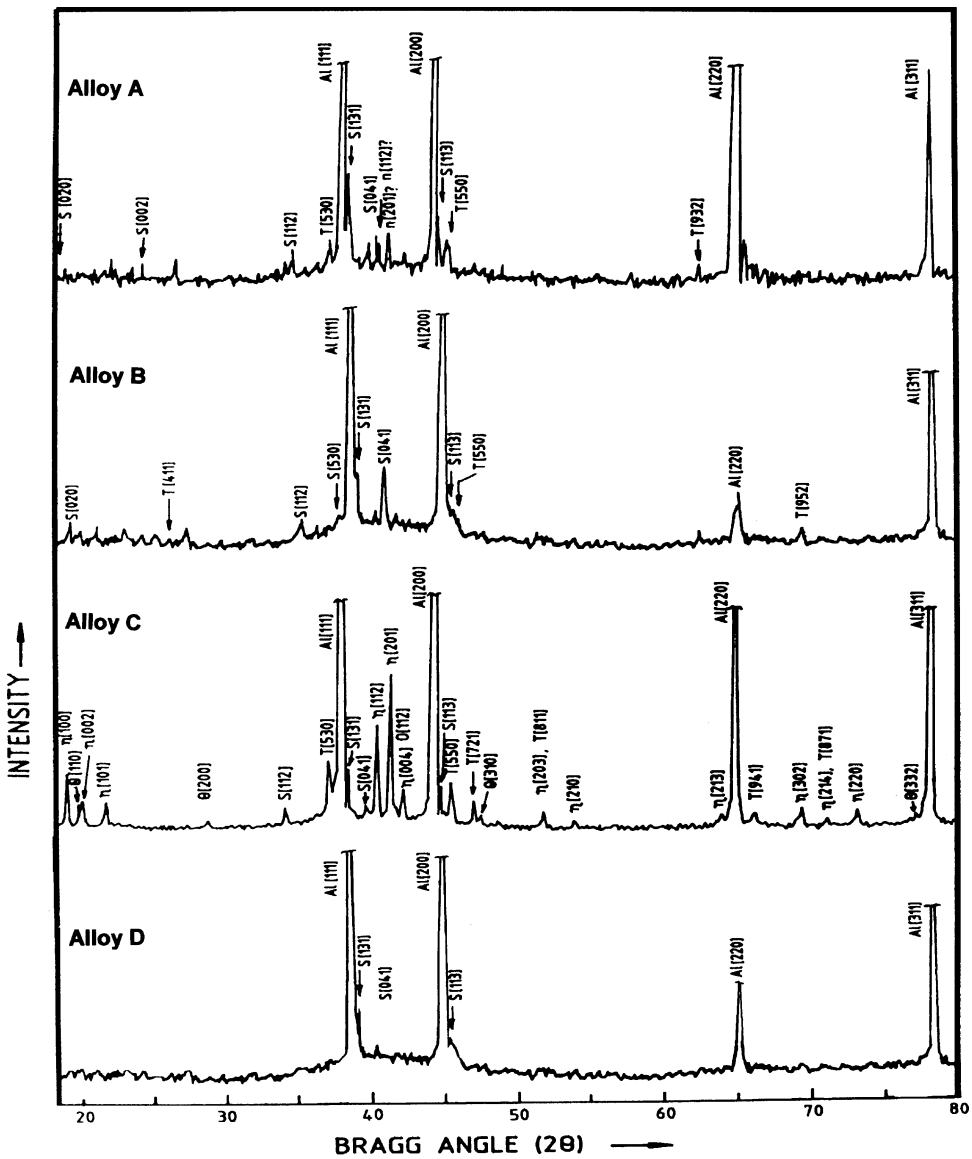

Fig. 8. XRD traces of annealed (35 h at 450 °C) 7055 aluminum alloys having 8 wt.% zinc, 0.16 wt.% zirconium and varying copper and magnesium contents (wt.%) i.e. alloy A containing 2.3 wt.% Mg and 2.6 wt.% Cu, alloy B containing 2.0 wt.% Mg and 2.3 wt.% Cu, alloy C containing 1.8 wt.% Mg and 2.6 wt.% Cu, and alloy D containing 1.8 wt.% Mg and 2.0 wt.% Cu.

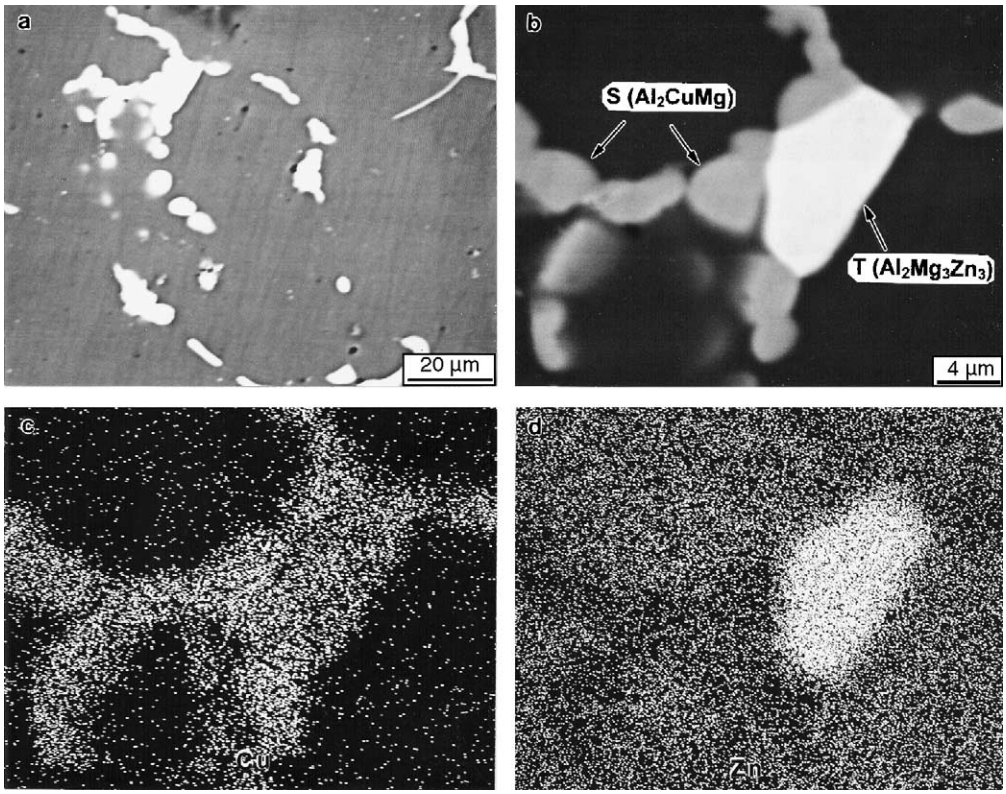

Fig. 9. BSE image obtained from annealed (35 h at 450 °C) Al–8.0Zn–2.0Mg–2.3Cu–0.16Zr alloy showing (a) typical annealed microstructure, (b) magnified image of S phase particles (marked by arrows and labeled) and T phase (marked by an arrow and labeled). (c) Cu and (d) Zn X-ray maps of the BSE image shown in (b).

Our studies showed that in addition to the insoluble Fe-bearing and Mg_2Si phases (that are present in all the alloys), varying proportions of alloy phases based on S(Al_2CuMg) and/or T($Al_2Mg_3Zn_3$) phases survived the homogenization treatment. The XRD traces of the annealed alloys are shown in Fig. 8. The XRD peaks could be identified due to α(Al) solid solution (labeled Al), $Al_2Mg_3Zn_3$ (labeled T), and Al_2CuMg (labeled S); the corresponding crystallographic planes are also marked in parentheses. Fig. 8 also shows that the presence of T and S phases could be detected in alloys A and B. Traces of T phase could also be detected in alloy C by EPMA, however, their volume fraction is too low to be detected by XRD. On the other hand, in alloy C and even in alloy D, small amounts of undissolved S phase could be detected.

Fig. 9(a) represents a scanning electron micrograph showing distribution of secondary alloy phases in alloy B after the annealing treatment. Fig. 9(b) shows a magnified image of secondary phases with enhanced contrast in BSE mode. EPMA analyses revealed the presence of T (brighter contrast) and S phases (dark contrast) and it corroborates well with the XRD results.

Table 3 shows the chemical compositions of the secondary phases that survived the homogenizing annealing treatment. Quantitative WDS analyses showed that the T phase dissolves similar amounts of copper as in the as-cast condition. Whilst, the chemical composition of the S phase is close to that of the equilibrium S phase. This implies that zinc diffused out of the S phase during the annealing treatment. Copper and zinc X-ray maps from T and S phases as shown in Fig. 9(c) and (d), respectively further support such an interpretation.

Table 3
Chemical compositions (wt.%) of alloy phases present in annealed alloys

Alloy	Phases present	Zn	Mg	Cu	Al
A	T-base	37.6	21.1	26.0	15.3
	S-base	3.2	17.1	39.4	40.3
B	T-base	35.1	19.9	24.8	20.2
	S-base	2.7	16.3	36.1	44.9
C	T-base	33.5	20.6	25.8	20.1
	S-base	2.9	16.0	34.1	47.0
D	S-base	3.2	16.5	39.7	40.6
E	T-base	37.4	20.3	25.8	16.5

4. Summary and conclusions

1. The major secondary alloy phases present in the as-cast microstructure of 7055 aluminum alloys are based on η($MgZn_2$), T($Al_2Mg_3Zn_3$), S(Al_2CuMg), and θ(Al_2Cu). The T phase dissolves copper in the range of 27–28 wt.%. The S phase is found to be associated with zinc in the

range of 25–30 wt.%. The θ phase does not dissolve either zinc or magnesium in agreement with the previously documented results.

2. Increasing Cu:Mg ratio in the alloy composition has the effect of decreasing the lattice parameter of the T phase.
3. Below the average copper and magnesium contents in the alloy composition i.e. Al–8.0Zn–2.0Mg–2.3Cu–0.16Zr, the alloys could be homogenized substantially using a commercially viable homogenizing annealing treatment i.e. 35 h at 450 °C. The undissolved alloy phases, still remaining in alloys after the homogenizing annealing treatment, are a combination of Cu-bearing T and Zn-free S. The latter implies that zinc diffuses out of the S phase during annealing.

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